

Impact of Quantization and Structured Pruning on Battery SOH Prediction Models across Architectures

Anonymous

Anonymous,

Abstract

- Core objective: investigate how pruning, quantization, and knowledge distillation affect SOH prediction quality and embedded deployment cost across LSTM, GRU, CNN, and TCN architectures.
- Study principle: architectures are trained under a harmonized protocol first, then optimized under controlled compression settings to enable fair cross-architecture comparison.
- Evaluation focus: MAE/RMSE on SOH prediction and embedded KPIs (flash, RAM, latency, energy).
- Expected outcome: architecture-specific design guidance for resource-constrained BMS implementations.

Keywords: Battery SOH prediction, Structured pruning, Quantization, Knowledge distillation, Embedded AI, LSTM, GRU, CNN, TCN, BMS

1. Introduction

- Battery management systems require accurate SOH prediction under strict constraints on flash, RAM, latency, and energy.
- Many published studies emphasize predictive accuracy but report only limited evidence for embedded deployment trade-offs.

- 23 • This paper focuses on architecture-dependent optimization behavior and asks
24 whether different model families benefit equally from pruning, quantization,
25 and distillation.
- 26 • The target is a comparable SOH-accuracy range across architectures before
27 compression, followed by systematic compression analysis.
- 28 • Figure 1 (planned, position in this section): high-level study concept showing
29 the flow from baseline models to compression methods and final embedded
30 evaluation.

31 2. Related Work and Research Gap

- 32 • Existing SOH literature covers recurrent and convolutional sequence models,
33 but direct cross-architecture compression comparisons are still limited.
- 34 • Most compression studies are model-centric and do not combine battery-
35 domain evaluation with embedded KPIs in one unified protocol.
- 36 • The open gap is a controlled cross-architecture analysis that links compression
37 decisions directly to BMS-relevant engineering costs.
- 38 • Figure 2 (planned, position in this section): literature positioning map con-
39 trasting accuracy-centric studies, compression-centric studies, and the pro-
40 posed integrated approach.

41 3. Methodology

42 3.1. Comparison Objective and Fairness Principle

- 43 • Step 1: establish strong baseline models per architecture in a comparable
44 SOH-accuracy band.
- 45 • Step 2: apply optimization methods under controlled settings and evaluate
46 accuracy-cost shifts.

47 • Step 3: compare architecture-specific robustness and derive deployment-oriented
48 recommendations.

49 • Fairness rule: model sizes may differ when required to reach comparable base-
50 line quality, but data split, metric definition, and benchmark protocol remain
51 fixed.

52 *3.2. Data, Target, and Split Strategy*

53 • Prediction target: battery SOH over aging trajectories.

54 • Feature basis from current code: Voltage, Current, Temperature, EFC, and
55 Q_c.

56 • Feature preprocessing: hourly aggregation into mean, std, min, and max
57 channels.

58 • Common partitioning: train cells C01/C03/C05/C09/C13/C17/C25/C27/C29;
59 validation cells C07/C15/C21; test cells C11/C19/C23.

60 • Figure 3 (planned, position in this subsection): SOH trajectories of all cells
61 with color-coded train/validation/test assignment.

62 This split figure documents partition coverage and helps verify that all sets
63 contain representative SOH evolution ranges.

64 *3.3. Architecture Portfolio*

65 • LSTM reference model: version 0.1.2.3, hidden size 192, 3 recurrent layers.

66 • GRU reference model: version 0.3.1.1, hidden size 256, 4 recurrent layers.

67 • TCN reference model: version 0.2.2.1, hidden size 96 with causal dilation
68 stack.

69 • CNN reference model: version 0.4.1.1, hidden size 224 with dilated causal
70 blocks.

71 • Figure 4 (planned, position in this subsection): architecture schematic panel
72 with one compact diagram per model family.

Table 1: Architecture comparison for the SOH optimization study (baseline FP32 models).

Family	Version	Hidden Size	Main Depth	Parameters	Key Structure
LSTM	0.1.2.3	192	3 LSTM layers	1,041,185	LSTM-Hybrid-Seq2Seq with residual connections
GRU	0.3.1.1	256	4 GRU layers	1,919,617	GRU-Hybrid-Seq2Seq with residual connections
TCN	0.2.2.1	96	4 blocks, dilations 1/2/4/8/16	266,977	Causal TCN Seq2Seq
CNN	0.4.1.1	224	4 dilated causal blocks	2,056,929	Dilated causal CNN Seq2Seq

3.4. Training and Evaluation Protocol

- Training setup is harmonized across architectures (seed 42, seq2seq objective, smoothness regularization, cosine warm restarts, early stopping, gradient clipping).
- Main predictive metrics are MAE and RMSE; optional analyses can include EOL-oriented indicators.
- Current practical workflow in the project: one main run per selected model version, with model acceptance based on test MAE in a practically usable range.
- Figure 5 (planned, position in this subsection): training/evaluation flow chart from data loading to baseline model selection.

3.5. Model Optimization Methods

- Pruning: structured pruning is applied in percentage steps to quantify architecture-specific sensitivity and breakpoints.
- Quantization: precision is reduced from FP32 to lower-bit formats to evaluate memory and runtime savings versus accuracy loss.
- Knowledge distillation: teacher-student training is integrated to recover predictive quality of compressed models.
- Combined settings: selected model variants are evaluated with joint optimization (for example pruning + quantization, with optional distillation support).

- Figure 6 (planned, position in this subsection): optimization pipeline diagram showing baseline model, individual optimization branches, and combined branch.

3.6. *Embedded Benchmark Protocol*

- All variants are benchmarked with the same target platform and firmware toolchain.
- Inference is performed with batch size 1 and fixed warm-up, because deployment is stateful and sample-by-sample in online BMS operation.
- Embedded KPIs: MAE, flash, RAM, latency (mean and std), and energy.
- Quantization path in current code: dynamic INT8 for LSTM/GRU, static FX INT8 for CNN, and stabilized head-focused static FX for TCN.
- Figure 7 (planned, position in this subsection): benchmark setup including host pipeline, firmware execution, and metric logging flow.

4. Results

4.1. *Baseline Model Comparison*

- Report baseline MAE/RMSE per architecture on the shared test set.
- Compare baseline model sizes and baseline embedded costs before any compression.
- Figure 8 (planned, position in this subsection): grouped bars for baseline MAE/RMSE and baseline footprint.

4.2. *Pruning Results*

- Report MAE/RMSE trends over pruning levels per architecture.
- Report flash/RAM/latency/energy changes over pruning levels.

116 • Identify pruning thresholds where performance leaves the acceptable operat-
117 ing range.

118 • Figure 9 (planned, position in this subsection): pruning sensitivity curves and
119 per-architecture Pareto view.

120 *4.3. Quantization Results*

121 • Compare FP32 and lower precision variants per architecture.

122 • Quantify trade-offs between prediction quality and embedded resource gains.

123 • Distinguish robust quantization cases from degradation-prone cases.

124 • Figure 10 (planned, position in this subsection): precision-vs-MAE and precision-
125 vs-resource multi-panel plot.

126 *4.4. Distillation and Combined Optimization Results*

127 • Evaluate whether distillation recovers accuracy for compressed student mod-
128 els.

129 • Compare single-method and combined-method variants within one decision
130 space.

131 • Analyze additive versus interaction effects between optimization methods.

132 • Figure 11 (planned, position in this subsection): combined optimization scat-
133 ter with non-dominated configurations.

134 *4.5. Cross-Architecture Decision View*

135 • Summarize all accepted variants in architecture-wise and global Pareto fronts.

136 • Highlight architecture-dependent operating regions for low-memory, low-latency,
137 and low-energy constraints.

138 • Figure 12 (planned, position in this subsection): final decision map for archi-
139 tecture and optimization selection.

140 5. Discussion

- 141 • Interpret why architectures react differently to identical optimization settings.
- 142 • Discuss where higher model complexity is justified by robustness gains and
143 where lightweight structures dominate.
- 144 • Relate results to practical BMS deployment constraints and integration risks.
- 145 • Clarify methodological limits (data regime, target platform scope, and con-
146 figuration breadth).
- 147 • Figure 13 (planned, position in this section): qualitative decision matrix link-
148 ing constraints to recommended architecture-method combinations.

149 6. Conclusion and Outlook

- 150 • The study provides a controlled architecture-level comparison of pruning,
151 quantization, and distillation for SOH prediction.
- 152 • The central contribution is a reproducible accuracy-cost analysis that directly
153 supports embedded design decisions.
- 154 • Future work extends the study to additional chemistries, broader operating
155 regimes, and tighter hardware-in-the-loop integration.

156 Figure Directory (Working Draft)

- 157 • Figure 1: Study concept from baseline training to compression and embedded
158 evaluation.
- 159 • Figure 2: Literature positioning map and research gap framing.
- 160 • Figure 3: SOH trajectories with train/validation/test split highlighting.
- 161 • Figure 4: Architecture panel for LSTM, GRU, CNN, and TCN.
- 162 • Figure 5: Training and baseline-selection workflow.

- 163 • Figure 6: Optimization workflow (pruning, quantization, distillation, com-
164 bined branch).
- 165 • Figure 7: Embedded benchmark setup and metric acquisition flow.
- 166 • Figure 8: Baseline MAE/RMSE and footprint comparison.
- 167 • Figure 9: Pruning sensitivity and pruning Pareto view.
- 168 • Figure 10: Quantization sensitivity and precision-resource trade-off.
- 169 • Figure 11: Distillation and combined optimization scatter with non-dominated
170 points.
- 171 • Figure 12: Cross-architecture decision map.
- 172 • Figure 13: Discussion matrix linking deployment constraints to model-method
173 choices.

174 **Data availability (placeholder)**

- 175 • Dataset DOI, code repository links, and benchmark scripts will be inserted
176 in the final manuscript version.